LA HW

Included exercises: 1-2, 30-31, 38, 49, 51, 53

Exercises 1-8

In Exercises 1–8, determine whether each linear transformation is one-to-one.

Original Exercise 1

-to-one.

$T: \mathcal{M}_{2\times 2} \to \mathcal{M}_{2\times 2}$ is the linear transformation defined by $T(A) = A \begin{bmatrix} 1 & 2 \end{bmatrix}$

Original Exercise 2

by $T(A) = A \begin{bmatrix} 3 & 4 \end{bmatrix}$.

$U: \mathcal{M}_{2\times 2} \to \mathcal{R}$ is the linear transformation defined by $U(A) = \operatorname{trace}(A)$.

Exercises 30-37

In Exercises 30–37, determine whether each transformation T is linear. If T is linear, determine if it is an isomorphism. Justify your conclusions.

Original Exercise 30

conclusions.

$T: \mathcal{M}_{n\times n} \to \mathcal{R}$ defined by $T(A) = \det A$

Original Exercise 31

$T: \mathcal{P} \to \mathcal{P}$ defined by $T(f(x)) = xf(x)$

Original Exercise 38

Let $S = \{s_1, s_2, \ldots, s_n\}$ be a set consisting of n elements, and let $T \colon \mathcal{F}(S) \to \mathcal{R}^n$ be defined by

$$T(f) = \begin{bmatrix} f(s_1) \\ f(s_2) \\ \vdots \\ f(s_n) \end{bmatrix}.$$

Prove that T is an isomorphism.

Original Exercise 49

Let N denote the set of nonnegative integers, and let V be the subset of $\mathcal{F}(N)$ consisting of the functions that are zero except at finitely many elements of N.

(a) Show that V is a subspace of $\mathcal{F}(N)$.

(b) Show that V is isomorphic to $\mathcal{P}$. *Hint:* Choose $T \colon V \to \mathcal{P}$ to be the transformation that maps a function f to the polynomial having $f(i)$ as the coefficient of x^i.

Original Exercise 51

answer.

Let V be the subset of $\mathcal{P}_4$ of polynomials of the form $ax^4 + bx^2 + c$, where a, b, and c are scalars.

(a) Prove that V is a subspace of $\mathcal{P}_4$.

(b) Prove that V is isomorphic to $\mathcal{P}_2$ by constructing an

Exercise 53

The inverses of T, U, and U
We summarize these observ

Original Exercise 53

Let V, W, and Z be vector spaces and $T: V \to W$ and